

MEHDI MECHKAK

Senior Applied AI & Systems Engineer | RAG, FastAPI, n8n & Industrial Control

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PROFESSIONAL SUMMARY

Applied AI & Systems Engineer with dual expertise in production LLM architectures (RAG, FastAPI microservices, n8n webhook pipelines, pgvector) and heavy industrial control systems (SCADA, PLC, electrical engineering). Currently pursuing a Master's degree in Embedded Systems and Autonomous Systems at Université Paris-Saclay. Bilingual in English (C2) and French (C1/C2).

ACADEMIC EDUCATION

Master's Degree (MSc) — Embedded Systems & Autonomous Systems (SESI) 2026 – 2027
Université Paris-Saclay, Orsay, France

- Advanced specialization in Hardware/Software Co-design, Real-time Embedded Operating Systems, Autonomous Robotics Control & Embedded AI Security.

Bachelor's Degree (Licence 3) — Electronics, Electrical Engineering & Automation (EEA) 2025 – 2026
Université Paris-Saclay, Orsay, France

- Core Focus: Electronics, Signal Processing, Control Systems, Regulation, C Programming & Computer Networks.

Cycle Ingénieur (Engineering Program) — Electrical Engineering 2023 – 2025
National Polytechnic School (ENP), Algiers, Algeria

- Core Focus: High-Voltage Systems, Power Transformers, Gas Turbines, Industrial Automation & Power Distribution.

Classes Préparatoires aux Grandes Écoles (Preparatory Cycle) 2021 – 2023
National Polytechnic School (ENP), Algiers, Algeria

- Intensive two-year engineering prep in Advanced Mathematics, Physics, Signal Processing, and Computer Science.

PRODUCTION AI & SYSTEMS ENGINEERING PROJECTS

Hotel Concierge AI (ConciergeFlow) — AI Systems Engineer Jan 2024 – Jun 2024
Live Demo: <https://flowarchitect-agency.github.io/hotel-concierge-ai/#demo>

- Architected 24/7 conversational AI concierge for Paris luxury boutique hotels providing sub-2.0s responses on web chat and WhatsApp.
- Constructed n8n workflow pipelines integrated with Airtable APIs for live room lookups and spa upsell recommendations.
- Built automated regression test harness in Python verifying response times under 2 seconds and 100% grounded inventory answers.

Polylingual AI Support Agent — Applied AI Engineer & Architect Mar 2024 – Aug 2024
Live Demo: <https://d15fb7qio4xmdn.cloudfront.net>

- Designed evidence-first French/English support automation platform using dual hybrid lexical + pgvector semantic retrieval.
- Implemented 78% confidence gating mechanism for human escalations, achieving an **88% auto-resolution rate**.
- Provisioned AWS infrastructure via Terraform (ECS Fargate, RDS PostgreSQL, CloudFront) with Prometheus/Grafana telemetry.

AI Internal Tool Builder (Nexus Control Room) — Full-Stack Engineer & AI Architect Jun 2024 – Nov 2024
GitHub: <https://github.com/FlowArchitect-Agency/ai-internal-tool-builder>

- Built natural language workflow node generator converting EN/FR prompts into visual node maps in <2.1 seconds.
- Enforced atomic server-side state machine approval gates (PENDING, APPROVED, RUNNING) in Next.js 14, Prisma 7, and PostgreSQL.

Automated BI Pipeline with AI Enrichment — Data Engineer & AI Specialist Aug 2024 – Present
GitHub: <https://github.com/FlowArchitect-Agency/automated-bi-pipeline>

- Orchestrated daily Apache Airflow ETL DAG processing raw data across 5 e-commerce channels (€4.97M+ revenue, 20,415 orders).
- Enriched support tickets and reviews using LLM sentiment classification into PostgreSQL data mart powering Streamlit BI dashboards & WeasyPrint PDF reports.

INDUSTRIAL CONTROL & ELECTRICAL ENGINEERING EXPERIENCE

Graduate Engineering Intern — Power Systems & Protection
SONATRACH, Equinor, ENI (Oil & Gas) — Algeria

2024 – 2025

- Conducted detailed analysis of Gas Turbine Generators (GTG) and high-voltage protection systems for 10 kV motors using Megger testing equipment.

Graduate Engineering Intern — Electrical & Instrumentation
SONATRACH — CEPSA (Oil & Gas) — Algeria

2023 – 2024

- Maintained UPS systems, transformers, and rotating machinery. Monitored and optimized Distributed Control Systems (DCS) and PLC logic.

TECHNICAL SKILLS & LANGUAGES

AI & Backend: Python, FastAPI, pgvector, RAG, Next.js 14, TypeScript, n8n, Node.js, SQL, PostgreSQL, Apache Airflow, Docker, Terraform, AWS.

Engineering & Code: C, C++, Java, MATLAB, Signal Processing, SCADA, PLC (Siemens/Schneider), DCS, HMI, Modbus, OPC-UA, LTSpice, AutoCAD.

Languages & Certs: English (C2 Bilingual / EF SET Certified), French (C1/C2 Fluent), Arabic (C2 Native). Wind Energy & Green Hydrogen Training (Univ. Rostock).